

Course Topics

Table of contents

1	BUS 411 — Fixed Income	2
1.1	Lecture Topics	2
1.2	Lecture 1: History and Role of Fixed Income Markets	3
1.2.1	1.1 What is fixed income?	3
1.2.2	1.2 Historical development of fixed income markets	3
1.2.3	1.3 Economic role of fixed income securities	3
1.3	Lecture 2: Time Value of Money and Discounting	3
1.3.1	2.1 Time value of money	3
1.3.2	2.2 Compounding conventions	3
1.3.3	2.3 Discounting fixed income cash flows	4
1.4	Lecture 3: Annuities and Liability Valuation	4
1.4.1	3.1 Fixed annuities	4
1.4.2	3.2 Variable annuities	4
1.4.3	3.3 Defined benefit pension plans	4
1.4.4	3.4 Liability valuation	4
1.5	Lecture 4: Government Bonds	5
1.5.1	4.1 Types of government securities	5
1.5.2	4.2 Features of government bonds	5
1.5.3	4.3 Valuation of government bonds	5
1.5.4	4.4 Government bond markets	5
1.6	Lecture 5: Corporate Bonds and Credit Risk	5
1.6.1	5.1 Corporate bond features	5
1.6.2	5.2 Valuation of corporate bonds	6
1.6.3	5.3 Credit risk	6
1.6.4	5.4 Corporate bond market considerations	6
1.7	Lecture 6: Yield Measures and the Term Structure of Interest Rates	6
1.7.1	6.1 Yield measures	6
1.7.2	6.2 Spot and forward rates	6
1.7.3	6.3 Yield curve construction and interpretation	6

1.7.4	6.4	Term structure concepts	7
1.8		Lecture 7: Duration	7
1.8.1	7.1	Interest rate risk in fixed income	7
1.8.2	7.2	Macaulay duration	7
1.8.3	7.3	Modified duration	7
1.8.4	7.4	Determinants of duration	7
1.9		Lecture 8: Convexity	8
1.9.1	8.1	Limits of duration-only approximation	8
1.9.2	8.2	Convexity measure	8
1.9.3	8.3	Duration-convexity approximation	8
1.9.4	8.4	Comparing bonds using convexity	8
1.10		Lecture 9: Fixed Income Risk Management	8
1.10.1	9.1	Duration-based risk management	8
1.10.2	9.2	Immunization	8
1.10.3	9.3	Convexity in risk management	9
1.10.4	9.4	Practical limitations	9
1.11		Lecture 10: Bonds with Contingencies	9
1.11.1	10.1	Bonds with embedded options	9
1.11.2	10.2	Valuation issues for contingent bonds	9
1.11.3	10.3	Mortgage-backed and prepayable securities	9
1.11.4	10.4	Risk characteristics	10
1.12		Lecture 11: Fixed Income Derivative Securities	10
1.12.1	11.1	Overview of fixed income derivatives	10
1.12.2	11.2	Interest rate forwards and futures	10
1.12.3	11.3	Interest rate swaps	10
1.12.4	11.4	Options on interest rates	10
1.13		Coverage of Core Course Themes	10
1.13.1	1.	History of fixed income securities	11
1.13.2	2.	Valuation methods for fixed income securities	11
1.13.3	3.	Fixed income risk management using duration and convexity	11

1 BUS 411 — Fixed Income

1.1 Lecture Topics

This document lists the planned lecture topics and subtopics for the course.

1.2 Lecture 1: History and Role of Fixed Income Markets

1.2.1 1.1 What is fixed income?

- Definition of fixed income securities
- Debt versus equity financing
- Main issuers and investors

1.2.2 1.2 Historical development of fixed income markets

- Early sovereign borrowing
- Development of government debt markets
- Growth of corporate bond markets

1.2.3 1.3 Economic role of fixed income securities

- Funding governments
 - Funding corporations
 - Saving, retirement, and liability management
-

1.3 Lecture 2: Time Value of Money and Discounting

1.3.1 2.1 Time value of money

- Present value
- Future value
- Discount factors

1.3.2 2.2 Compounding conventions

- Annual compounding
- Semiannual compounding
- Continuous compounding

1.3.3 2.3 Discounting fixed income cash flows

- Cash flow timelines
 - Pricing as present value of future cash flows
 - Relation between discount rates and value
-

1.4 Lecture 3: Annuities and Liability Valuation

1.4.1 3.1 Fixed annuities

- Ordinary annuities
- Annuities due
- Perpetuities

1.4.2 3.2 Variable annuities

- Basic structure
- Investment-linked cash flows
- Valuation considerations

1.4.3 3.3 Defined benefit pension plans

- Nature of pension liabilities
- Discounting promised payments
- Funding and valuation issues

1.4.4 3.4 Liability valuation

- Present value of liabilities
 - Matching assets to liabilities
 - Liability sensitivity to interest rates
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1.5 Lecture 4: Government Bonds

1.5.1 4.1 Types of government securities

- Treasury bills
- Treasury notes
- Treasury bonds

1.5.2 4.2 Features of government bonds

- Face value
- Coupon rate
- Maturity
- Clean price and dirty price

1.5.3 4.3 Valuation of government bonds

- Pricing from discounted cash flows
- Yield to maturity
- Accrued interest

1.5.4 4.4 Government bond markets

- Primary market
 - Secondary market
 - Benchmark role of sovereign debt
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1.6 Lecture 5: Corporate Bonds and Credit Risk

1.6.1 5.1 Corporate bond features

- Secured and unsecured bonds
- Seniority
- Covenants

1.6.2 5.2 Valuation of corporate bonds

- Pricing with promised cash flows
- Credit spread over government rates
- Yield spread interpretation

1.6.3 5.3 Credit risk

- Default risk
- Recovery risk
- Credit ratings

1.6.4 5.4 Corporate bond market considerations

- Liquidity
 - Issuer quality
 - Term to maturity and risk
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1.7 Lecture 6: Yield Measures and the Term Structure of Interest Rates

1.7.1 6.1 Yield measures

- Current yield
- Yield to maturity
- Holding period return

1.7.2 6.2 Spot and forward rates

- Spot rates
- Forward rates
- Discount function

1.7.3 6.3 Yield curve construction and interpretation

- Normal, flat, and inverted curves
- Economic interpretation
- Expectations about future rates

1.7.4 6.4 Term structure concepts

- Expectations view
 - Liquidity preference
 - Market segmentation intuition
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1.8 Lecture 7: Duration

1.8.1 7.1 Interest rate risk in fixed income

- Bond price and yield relationship
- Why prices fall when yields rise

1.8.2 7.2 Macaulay duration

- Definition
- Weighted average time to cash flow receipt
- Interpretation

1.8.3 7.3 Modified duration

- Definition
- Approximate price sensitivity
- Duration as first-order risk measure

1.8.4 7.4 Determinants of duration

- Coupon
 - Maturity
 - Yield level
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1.9 Lecture 8: Convexity

1.9.1 8.1 Limits of duration-only approximation

- Nonlinearity in bond prices
- Curvature of the price-yield relationship

1.9.2 8.2 Convexity measure

- Definition
- Interpretation
- Why convexity matters

1.9.3 8.3 Duration-convexity approximation

- First-order and second-order effects
- Improved bond price change estimates
- Application to larger yield changes

1.9.4 8.4 Comparing bonds using convexity

- Price sensitivity differences
 - Trade-off between yield and convexity
 - Portfolio implications
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1.10 Lecture 9: Fixed Income Risk Management

1.10.1 9.1 Duration-based risk management

- Measuring portfolio interest rate exposure
- Portfolio duration

1.10.2 9.2 Immunization

- Asset-liability matching
- Duration matching
- Protecting against small parallel shifts in yields

1.10.3 9.3 Convexity in risk management

- Why duration matching alone is not enough
- Role of convexity in immunization
- Rebalancing issues

1.10.4 9.4 Practical limitations

- Non-parallel yield curve shifts
 - Model risk
 - Transaction costs
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1.11 Lecture 10: Bonds with Contingencies

1.11.1 10.1 Bonds with embedded options

- Callable bonds
- Puttable bonds
- Convertible features as a general idea

1.11.2 10.2 Valuation issues for contingent bonds

- Cash flows depend on future events
- Option-like features in bond valuation
- Impact on price and yield

1.11.3 10.3 Mortgage-backed and prepayable securities

- Prepayment risk
- Negative convexity
- Valuation challenges

1.11.4 10.4 Risk characteristics

- Reinvestment risk
 - Call risk
 - Extension and contraction risk
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1.12 Lecture 11: Fixed Income Derivative Securities

1.12.1 11.1 Overview of fixed income derivatives

- Why derivatives are used
- Hedging, speculation, and portfolio management

1.12.2 11.2 Interest rate forwards and futures

- Basic contract structure
- Hedging interest rate exposure
- Relation to underlying fixed income securities

1.12.3 11.3 Interest rate swaps

- Fixed-for-floating exchange
- Basic valuation intuition
- Uses in risk management

1.12.4 11.4 Options on interest rates

- Caps
 - Floors
 - Basic role in managing rate risk
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1.13 Coverage of Core Course Themes

The course is organized around three major themes:

1.13.1 1. History of fixed income securities

- Covered primarily in Lecture 1

1.13.2 2. Valuation methods for fixed income securities

- Fixed and variable annuities: Lecture 3
- Government bonds: Lecture 4
- Corporate bonds: Lecture 5
- Defined benefit pension plans: Lecture 3
- Bonds with contingencies: Lecture 10
- Fixed income derivative securities: Lecture 11

1.13.3 3. Fixed income risk management using duration and convexity

- Duration: Lecture 7
- Convexity: Lecture 8
- Risk management and immunization: Lecture 9